

# Factoring Algebraic Expressions (Part II)

## Factoring the Difference of Two Squares

Key Concept:  $(x-3)(x+3) = x^2 + 3x - 3x - 9 = x^2 - 9$

- ①  $x^2 - 16 = (x-4)(x+4)$
- ②  $y^2 - 64 = (y+8)(y-8)$
- ③  $z^2 - 4 = (z-2)(z+2)$
- ④  $x^2 - 81 = (x-9)(x+9)$
- ⑤  $y^2 - 49 = (y-7)(y+7)$
- ⑥  $z^2 - 25 = (z+5)(z-5)$
- ⑦  $x^2 - 1 = (x-1)(x+1)$
- ⑧  $4y^2 - 9 = (2y+3)(2y-3)$
- ⑨  $9z^2 - 16 = (3z-4)(3z+4)$
- ⑩  $25x^2 - 36 = (5x-6)(5x+6)$
- ⑪  $49y^2 - 100 = (7y-10)(7y+10)$
- ⑫  $36z^2 - 1 = (6z-1)(6z+1)$
- ⑬  $81x^2 - 4 = (9x+2)(9x-2)$
- ⑭  $36 - x^2 = (6-x)(6+x)$
- ⑮  $64 - 9y^2 = (8-3y)(8+3y)$

$$⑯ 100 - 49z^2 = (10-7z)(10+7z)$$

$$⑰ 25 - 9x^2 = (5-3x)(5+3x)$$

$$⑱ 1 - 64y^2 = (1-8y)(1+8y)$$

$$⑲ 4 - 81z^2 = (2-9z)(2+9z)$$

$$• ⑳ -x^2 + 9 = 9 - x^2 = (3-x)(3+x)$$

$$㉑ -y^2 + 100 = 100 - y^2 = (10-y)(10+y)$$

$$㉒ -64 + z^2 = z^2 - 64 = (z-8)(z+8)$$

$$• ㉓ 81z^4 - 16 = (9z^2+4)(9z^2-4) = (9z^2+4)(3z-2)(3z+2)$$

Factoring Quadratic Expressions with Two Variables and Factoring the Difference of Two Squares with Two Variables.

$$㉔ x^2 + 8xy + 15y^2 = (x+3y)(x+5y)$$

$$㉕ a^2 + 2ab - 63b^2 = (a+9b)(a-7b)$$

$$㉖ x^2 - xy - 30y^2 = (x-6y)(x+5y)$$

$$㉗ a^2 + 29ab - 30b^2 = (a+30b)(a-b)$$

$$㉘ x^2 + 7xy + 12y^2 = (x+3y)(x+4y)$$

$$㉙ a^2 - 2ab + b^2 = (a-b)(a-b)$$

$$㉚ x^2 - 4xy - 45y^2 = (x-9y)(x+5y)$$

$$㉛ a^2 - 4ab + 3b^2 = (a-3b)(a-b)$$



$$\textcircled{32} 2g^2 - 5gh + 2h^2 = (2g-h)(g-2h)$$

$$\textcircled{33} 3x^2 + 10xz - 8z^2 = (3x-2z)(x+4z)$$

$$\textcircled{34} 4x^2 + 19xz + 12z^2 = (4x+3z)(x+4z)$$

$$\textcircled{35} 5g^2 - 11gh + 2h^2 = (5g-h)(g-2h)$$

$$\textcircled{36} 3g^2 + 13gh - 10h^2 = (3g-2h)(g+5h)$$

$$\textcircled{37} 4a^2 - 20ab + 9b^2 = (2a-b)(2a-9b)$$

$$\textcircled{38} t^2 - 9r^2 = (t+3r)(t-3r)$$

$$\textcircled{39} 16a^2 - 25b^2 = (4a-5b)(4a+5b)$$

$$\textcircled{40} z^2 - y^2 = (z+y)(z-y)$$

$$\textcircled{41} 4r^2 - t^2 = (2r-t)(2r+t)$$

$$\textcircled{42} 36a^2 - 49b^2 = (6a-7b)(6a+7b)$$

$$\textcircled{43} 4g^2 - 81h^2 = (2g-9h)(2g+9h)$$

$$\textcircled{44} a^2 - 100b^2 = (a-10b)(a+10b)$$

$$\textcircled{45} 9x^2 - 4y^2 = (3x-2y)(3x+2y)$$

$$\textcircled{46} 49a^2 - 81b^2 = (7a-9b)(7a+9b)$$